

Branch: Computer Science and Engineering

Course Title: Computer Organization and Architecture **Course Code:** PCC CS-401

Total Marks: 200

Internal Marks: Theory-50 Practical-50

Class: B.Tech. 4th Sem.

Credits: 4 **L-T-P:** 3-0-2

External Marks: 100

Course Objectives:

- To provide knowledge on overview of IAS computer function and addressing modes.
- Hardware and software implementation of arithmetic unit to solve addition, subtraction, multiplication and division.
- To provide knowledge of memory technologies, interfacing techniques and sub system devices.

Course Outcome:

- Provide fundamentals on machine instructions and addressing modes.
- Comprehend the various algorithms for computer arithmetic.
- Analyze the performance of various memory modules in memory hierarchy.
- Compare and contrast the features of I/O devices and parallel processors.
- Outline the evaluation of memory organization.

Unit-I: Introduction - Computer architecture and computer organization, classes of computers, basic structure of computers. Basic Performance Equation, Clock Rate, Performance Measurement, Amdahl's law. Number system and representation of information, arithmetic and logical operation, hardware implementation, Real numbers - fixed and floating point, IEEE754 representation

Unit-II: Processor and Control Unit Design - Register Organization, Stack Organization, Instruction Set, Instruction formats, Instruction cycle, Types of operands, Addressing Modes. CPU Organization, Single bus organization, Multiple Bus organization, Control step sequence, Control unit functions, Hardwired control unit Design, Micro programmed control unit design.

Unit-III: I/O unit and Memory Organization - I/O Unit: Synchronous vs. Asynchronous I/O, I/O techniques - interrupts, polling, DMA, IOP.

Memory Organization: Characteristics and hierarchy of memory systems. Main Memory- ROM, Cache Memory, organization and Mapping, Replacement Algorithms. Associative Memories, Virtual Memory, Main Memory allocation, Segmentation, Paging. Secondary Storage: Magnetic Hard Disk, RAID, Optical memory, CDROM, DVD.

Unit-IV: Multiprocessor Configurations - Flynn's classifications, Parallel processing concepts, Introduction to pipeline processing and pipeline hazard, design issues of pipeline architecture, Instruction pipelining. Introduction to RISC & CISC Architectures.

Books Recommended:

Text Books

1. Computer Organization and Embedded Systems by Carl Hamacher, Z. Vranesic, S. Zaky, and N. Manjikian.
2. Computer System Architecture by M. Mano.

References

1. Computer Organization and Architecture by William Stallings, Pearson, 8th Edition.

List of Practicals:

Course exercises in Computer Organization are based on the topics covered under Theory Syllabus. The students are required to perform the following list of practical.

1. Implement a C program to convert a Hexadecimal, octal, and binary number to decimal number vice versa.
2. Implement a C program to perform Binary Addition & Subtraction.
3. Implement a C program to perform Multiplication of two binary numbers.
4. Implement a C-program to perform Multiplication of two binary numbers (signed) using Booth's Algorithms.
5. Registers and Counter Designs.
6. Understanding the behaviour of ALU, RAM, STACK and PROCESSOR from working modules and the modules designed by the student as part of the experiment.
7. Arithmetic Logic Unit.
8. Memory Design.
9. Direct Mapped and Associative cache Designs.
10. CPU Design.